The QueryGraph Stack

A governed semantic lakehouse for agentic AI — five coordinated open-source components that let agents answer over enterprise data while *proving* what they did.

THE THESIS

"The answer used OSI metric **X**, resolved to Sail table **Y**, under capability **C** and `odrl:read`, emitting OpenLineage run **R** anchored by attestation **A** — and source **Z** was denied, with a receipt." The differentiator for enterprise AI is not a longer context window; it is this verifiable chain from question to answer. Denials are receipts, never errors.

THE COMPONENTS

Component	Role	Release
Grust	Backend-neutral property graph for Rust; full GQL/Cypher reads (Full39075: CALL subqueries, TVFs, shortestPath); a dozen stores including Sail DataFrames.	0.12.0 "Lobster"
TypeSec	Security in the type system: unforgeable Capability<P, R> proofs; TypeDID Ed25519 signed agent envelopes; framework guards, MCP gate, enforcement proxy.	0.12.0 "Torcello"
LakeCat	Rust-native Iceberg REST catalog with stock-client conformance; catalog state, Sail planning, receipts, and graph projection bound to one table transition; QueryGraph bootstrap bundles.	0.3.0 "Ocelot"
Sail (fork)	Spark-compatible lakehouse engine with a Cypher graph-query extension compiled into the SQL frontend.	branch grust
QueryGraph	The semantic layer, Rust + Python: Croissant, CDIF, DID, ODRL projections, OSI business semantics, RBAC+ODRL dual gate, OpenLineage + attestations.	0.4.0 "Sentinel"

GOSHAWK AND SENTINEL: THE INTEROPERABILITY RELEASES

Real Ed25519 across languages — Python signs, Rust verifies; Rust signs, Python verifies; same seed means the same `did:key`; tampering fails on either side. **HTTP /v1 API** with TypeDID envelope auth (path- and body-bound; 401s carry receipts). **MCP servers in both languages** — Claude, LangChain, PydanticAI, LlamaIndex, CrewAI connect with zero adapters. **A2A agent card** published identically by both implementations. **Official OpenLineage conformance** — events schema-validated, deterministic UUIDv5 run ids shared across languages. Sentinel (0.4.0) adds the **governed navigator loop**: question → semantic search → policy receipts → SQL plans over allowed sources only → any LLM (or the deterministic baseline) → signed envelope + lineage + attestation.

HELD EQUIVALENT BY TESTS

An executable cross-language contract: byte-identical navigator bundles, matching governance semantics in the multi-agent story, both crypto directions verified live, both CLIs' lineage events schema-valid, identical agent cards, and a live auth round-trip — 49 Python + 40 Rust tests, CI on every push.

Links: github.com/querygraph/{querygraph, qg-rust, qg-python, grust, typesec, lakecat} • lakehq/sail • books: qg-rust/docs/book & docs/guide • querygraph.ai